

LiteGrip

LiteGrip Product Manual

NEXFORM ROBOTICS

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Revision History

Version	Date	Revision
V1.2.0	2026-11-03	Restructured the chapters; installation and wiring now point to the Usage Tutorial
V1.1.0	2026-10-22	Restructured the manual for end users and added the assembly drawings
V1.0.0	2026-09-24	Created the LiteGrip Product Manual skeleton

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1 Preface

English | 简体中文
LiteGrip is an adaptive two-finger parallel gripper from NEXFORM ROBOTICS, designed for research and education, AI robot development, and lightweight industrial automation, with an effective stroke of 85.000 mm. The gripper communicates with the host computer through a USB-CAN adapter (classic CAN, 1 Mbps) and ships with a Python SDK and factory calibration data. It suits grasping, transferring, loading and unloading, sorting, and algorithm validation.
This manual covers the product's appearance, technical specifications, safety notices, selection guidance, and daily maintenance. Start with the Usage Tutorial to install, wire, and run the gripper for the first time. For the software interface, parameters, and communication protocol, see the Software Development Manual.

1.1 Hardware components

The standard delivery and the equipment you need to provide are listed below; the sales configuration list is the final authority:

Category	Item	Description
Standard delivery	LiteGrip gripper body	Includes the drive motor, transmission, fingers, and fingertips
Standard delivery	CAN terminating resistors	One 120 Ω resistor at each end of the bus
You provide	24 V DC power supply	Powers the motor; the motor outputs no torque when it is not connected
You provide	USB-CAN adapter	Connects the gripper to the host computer
You provide	Host computer	Linux with SocketCAN support in the kernel

The gripper body consists of the following parts:

Part	Description
Gripper body	Contains the drive motor, transmission, and guide rail
Finger (left / right)	Moves in parallel along the guide rail; per-finger stroke 42.500 mm
Fingertip	The part that contacts the workpiece directly; dimensions are in “Dimensions and mounting interface”

1.2 Units

The physical quantities in this manual and in the SDK use the following units:

Parameter	Unit	Description
Aperture / position	mm	Millimetres
Velocity	mm/s	Millimetres per second
Torque	N·m	Newton-metres
Gripping force	N	Newtons
Temperature	°C	Degrees Celsius
Voltage	V DC	Volts (DC)

“Aperture” in this document and the position p returned by the SDK are the same quantity: the true distance between the inner faces of the two fingers. At full closure `p = 0`; at full opening `p = 85.000`, and the per-finger stroke is 42.500 mm. It is not a motor angle and not a per-finger displacement; it is exactly what a caliper measures between the two fingertips.

1.3 Safety

1.3.1 Overview

This section describes the safety principles you must follow when using the LiteGrip gripper and its system. Before installation and use, you must read this manual carefully and strictly follow every passage that carries a

severity label. Gripper systems carry potential hazards such as gripping, collision, and falling; users must fully recognize the operating risks and use the SDK or control software only after training.

1.3.2 Safety symbols

Safety notes in this manual use the labels below; the graphical symbols on the product are the authority:

- Danger: a hazardous situation that may cause death, personal injury, or severe equipment damage.
- Warning: a hazardous situation that, if not avoided, could cause death, personal injury, or severe equipment damage.
- High temperature: hot parts, where contact may cause burns or equipment damage.
- Caution: a general warning that, if not avoided, may cause personal injury or equipment damage.

1.3.3 Safety precautions

Handling the gripper

1. Install the gripper and the electrical equipment as this manual requires. The mounting surface must be flat and rigid enough, and the fastening bolts must carry an even load.
2. Before first use, have a qualified person inspect the gripper and the protection system, and test the hardware emergency stop, the mechanical stop, and the enable state.
3. Before use, check that the equipment is intact and mounted securely, and confirm that the fingers' motion range holds no obstructions, no people, and no cables that could be pulled in.
4. **You must provide a hardware emergency stop loop whose action does not depend on CAN communication or host software; you must provide a mechanical stop as the last line of defense.**
5. Read the documentation before development and debugging, and make sure the target position, velocity, and gripping force are correct. For the first motion, use a low velocity and a conservative stroke.
6. **This product has no power-off self-locking.** After disable or abnormal power-down the motor no longer outputs a holding torque, and the gripper may move under gravity or an external force. If the application does not allow movement after power-off, you must add a separate mechanical holding device.
7. Before installation, maintenance, or cleaning, you must disconnect the 24 V power supply and confirm that power is off.
8. Unauthorized personnel must not modify this unit's calibration file, the motor registers, or the host-side limiting parameters.

Personnel safety

1. Users must read the product manual and know the safe operating procedure and how to handle errors.
2. While the equipment is running, the fingers may be waiting for a command even when they look stopped, so treat the gripper as still in motion.
3. Define the motion range of the gripper and the end tool and set up a safe working zone; no person may enter the gripper's motion range while it is running.

4. Inspect the gripper regularly to keep the mounting bolts and the fingertips from loosening.
5. In manual teaching the motor outputs no torque, and the fingers move under gravity or an external force. To stop at a position, exit zero gravity mode after you let go. Debugging and calibration must be supervised by a person on site, with the hardware emergency stop available.

Electrical safety

1. Do not connect or disconnect the CAN or power cables while power is applied; cut the 24 V supply before installation and maintenance.
2. Check the power polarity; reversing it can damage the equipment. Confirm the polarity with a multimeter before powering up.
3. Protect the unit from water and moisture. If water gets in, cut the power immediately and contact support.
4. The 24 V supply must provide overcurrent and short-circuit protection.
5. Confirm a reliable ground connection between the host computer and the power supply.

1.3.4 Responsibility and standards

The gripper can form a complete system with other equipment, and this document does not cover all of the design, installation, and operation of that complete system. The safety of the complete system depends on how it is integrated; users must carry out a risk assessment for the complete system under the laws and safety standards of their country or region and take the corresponding protective measures.
Starting to use this gripper means you have read, understood, and accepted all the terms of this manual and of the safety information. Users are responsible for their own actions and their consequences, and must use the gripper only for legitimate purposes. We accept no liability for personal injury, accidents, or property damage caused by violation of the usage requirements or by force majeure.

1.3.5 Risk assessment

A risk assessment must consider every potential contact between a person and the gripper under normal use and foreseeable misuse, for example:

- Fingers being pinched by the gripper fingers;
- Impact between the gripper and the robot arm end or the fixtures;
- A workpiece falling because it is not gripped securely or the load is out of range;
- Handling sharp or hot workpieces;
- The gripper losing its holding torque and the workpiece coming loose after power-off or disable;
- Software failure leaving the gripper to keep executing a motion command.

Through the risk assessment, users must judge whether the relevant hazards constitute an unacceptable risk and take the corresponding measures. The gripper is not suitable for people who lack the guidance of a qualified professional or who do not have full civil capacity.

2 Technical specifications

2.1 General specifications

Parameter name	Value
Product name	LiteGrip adaptive two-finger parallel gripper
Structure	Adaptive two-finger parallel gripper
Drive method	Direct motor drive
Motor model	DM-J4310-2EC (MIT mode)
Communication	Classic CAN, 1 Mbps
Operating voltage	24 V DC (nameplate rating)
Effective stroke	85.000 mm
Per-finger stroke	42.500 mm (effective stroke ÷ 2)
Working stroke	0 ~ 85.000 mm, the entire stroke is usable
Power-off self-locking	None (holding torque is lost when power is removed)
Total mass	0.537 kg (gripper assembly) / 0.785 kg (holder assembly)

2.2 Electrical specifications

Parameter name	Value
Rated voltage	24 V DC
Undervoltage protection point	15 V
Overvoltage protection point	32 V
No-load current	0.037 A
Rated current	3.1 A
Peak current	16.5 A
Control / feedback period	200 Hz

Parameter name	Value
Motor overtemperature protection point	100 °C
Communication dropout protection	Automatically exits enable after 0.4 s
Status feedback mode	Polled (responds after receiving a command frame); passive listening is not supported

2.3 Interface and power

The LiteGrip external interface consists of three sets of connections, and applying 24 V is the precondition for motor torque.

2.3.1 Interface overview

Connection	Description
24 V power supply	Powers the motor; the motor produces no torque when it is not connected
CAN bus	Connects the gripper to the USB-CAN adapter, including CAN_H and CAN_L
CAN terminating resistors	One 120 Ω resistor at each physical end of the bus

> **Note**: The terminating resistors are mandatory. The typical symptom when they are missing is communication that drops in and out.

2.3.2 Power supply

Item	Specification
Rated voltage	24 V DC
Protection trip points	Undervoltage 15 V, overvoltage 32 V
No-load current	0.037 A

Item	Specification
Rated / peak current	3.1 A / 16.5 A

With only the USB-CAN adapter connected, the gripper can communicate and report status, but it will not move, and it reports an undervoltage fault (error code 0x9). This is normal behavior, not a fault.
Estimating from the peak current of 16.5 A, the 24 V supply should be at least 400 W; typical operation (rated 3.1 A) needs about 75 W.

2.3.3 CAN bus

Item	Specification
Protocol	Classic CAN (CAN 2.0, not CAN FD)
Baud rate	1 Mbps
Data frame length	8 bytes
Terminating resistors	120 Ω at each end of the bus
Control mode	MIT
Communication timeout protection	Automatically disables after 0.4 s

> **Note**: CAN FD must be explicitly disabled. If the interface is opened in FD mode, communication fails even with the correct baud rate, and this looks just like “no 24 V connected”, so it is easily misdiagnosed as a hardware fault.
>
> **Note**: Feedback from this product is polled. The motor answers only after it receives a command frame; passively listening on the bus never yields a feedback frame.

For interface configuration commands, frame formats, and registers, see the protocol appendix of the Software Development Manual.

2.4 Mechanical and environment

Parameter name	Value
Coil overtemperature protection point	100 °C
Driver board overtemperature protection point	120 °C

Note: The IP rating and CE, RoHS and REACH are not certified. Contact us before using the gripper in an environment with dust, moisture, or cutting fluid.

2.5 Dimensions and mounting interface

The external dimensions are taken from the two assembly drawings. The assemblies share the 148 mm width and the same mounting holes; they differ mainly in the drive end, the fingers, and the mass.

2.5.1 Gripper assembly

Item	Dimension
Overall dimensions (width × depth × height)	148 × 65.5 × 176.5 mm
Motor diameter	Ø57 mm
Drive section height	60 mm
Mounting flange section height	30 mm
Body depth	61 mm
Finger length	86 mm
Aperture in the drawing	85 mm
Mass	0.537 kg

Dimensioned drawing: gripper assembly (PDF)

2.5.2 Holder assembly

Item	Dimension
Overall dimensions (width × depth × height)	148 × 101.6 × 169 mm
Motor diameter	Ø53 mm
Rear drive section height	63.2 mm
Body depth	71.6 mm
Side finger assembly length	134 mm
Centre finger width	26 mm
Mass	0.785 kg

Dimensioned drawing: holder assembly (PDF)

2.5.3 Mounting interface

Item	Specification
Mounting holes	2 × M3 threaded holes
Hole spacing	15 mm
Applies to	Gripper assembly, holder assembly

Note: These are nominal dimensions from the linked assembly drawings and exclude cables and connectors. The 3D model and the drawings are the authority; contact us for cable and connector specifications.

Note: The holder assembly has a different motor diameter from the gripper assembly, so the electrical and motion specifications in this manual do not apply to it unchanged. Contact us before use.

2.6 Selection guidance

First confirm applicability:

Suitable	Not suitable
Grasping, transferring, loading/unloading, and sorting of small and medium-sized rigid objects	High-temperature objects, sharp objects
	Applications requiring precise gripping force (before force calibration is complete)
	Applications requiring power-off holding (this product has no self-locking)
	Dusty, humid, or corrosive environments (IP rating not certified)

Confirm the following in order when selecting:

#	Check	Basis for judgment
1	Object size	Whether it falls within the working stroke 0 ~ 85.000 mm
2	Object weight	Confirm it against your own workpiece, contact surface and pose; contact us for the data
3	Is release after power loss acceptable	This product has no power-off self-locking; if it is not acceptable, add a mechanical holding device

3 Use and maintenance

3.1 Daily use

Grasping and closing

`grasp()` closes until it contacts an object and is obstructed, then holds a force. The criterion is whether the fingers stop moving, so gripping an object and reaching the stroke endpoint both count as a successful grasp;

read the position to tell the two apart. `close()` is different: it presses onto the closed-side stop, and a stall halfway (an object) reports `ok=False`.

> **Note**: The gripping force parameter has not been calibrated against a real load; do not rely on it before force calibration is complete. For a pure position grasp, call `close()`, which takes no force argument and closes until an object obstructs it.

Manual guidance

To hold the gripper at a new position, put it in zero gravity mode; the motor then outputs no torque and you can push the fingers by hand. Leaving the mode holds the position at the moment you exit it.

> **Note**: Zero gravity mode does not itself hold position; the fingers stay wherever you let go and do not return on their own. Leaving the mode performs no range check, so it will hold a position even beyond the calibrated endpoint. Manual guidance must be supervised by a person on site.

3.2 Maintenance

Routine checks

The intervals below are a recommended maintenance convention, not measured specifications, and are not a lifetime or reliability figure for this product:

Interval	Check
Every shift	Whether the fingers move smoothly, and whether there is any abnormal noise
Weekly	Whether the cables are damaged, and whether the connectors are loose
Monthly	Whether the fastening bolts are loose, and whether the fingers are worn

Cleaning and storage

- After cutting power, wipe the housing with a dry, soft cloth;
- Do not use organic solvents or corrosive cleaners;
- Do not wash with water;
- Before long-term storage, disconnect the 24 V supply and store the unit in a dry environment free of corrosive gases.

Recalibration

After any of the following, you must recalibrate:

- Replacing the gripper body or the motor;

- Replacing or repairing a finger;
- A collision or an overload;
- An abnormal position reading after a power cycle.

3.3 Troubleshooting

Symptom	Possible cause	Remedy
No communication at all	CAN interface not configured, or CAN FD not disabled	Reconfigure the interface with CAN FD disabled
	Terminating resistors missing	Fit one 120 Ω at each end of the bus
	Loose cable	Check the connectors
Status can be read but the gripper does not move	The 24 V supply is not connected	Connect 24 V (the most common cause)
	Motor not enabled	Check whether the enable step succeeded
	No valid calibration	Reload the calibration or run the calibration procedure
The gripper stops halfway through a motion	It gripped an object or a foreign body	Check for obstructions
	Overcurrent or overtemperature protection tripped	Stop and let it cool, then check the load
The gripper stops at an endpoint and will not move	The target is outside the calibrated range and is silently clamped	The SDK does not report an error; check the target position
Grasping always reports failure	The fingers are still moving at timeout, usually because they never touched the object	Confirm the object is within the stroke range
	The timeout is set too short	Increase the grasp timeout
Insufficient gripping force	Force control has not been calibrated	Do not rely on the force parameter; use `close()` or add a mechanical limit
Abnormal noise or jamming during motion	Foreign matter on the guide rail or mechanical damage	Cut power and inspect; contact service

Symptom	Possible cause	Remedy
Shuts down after running for a while	Overtemperature protection	Reduce the load or improve heat dissipation

Error code reference:

Error code	Meaning	Action
0x0	Disabled	Normal (not-enabled state)
0x1	Enabled	Normal
0x8	Overvoltage	Check whether the supply exceeds 32 V
0x9	Undervoltage	Check whether 24 V is connected
0xA	Overcurrent	Check for binding and for an excessive load
0xB	MOS overtemperature	Stop and let it cool
0xC	Coil overtemperature	Stop and let it cool
0xD	Communication loss	Check the CAN wiring and the terminating resistors
0xE	Overload	Reduce the load

Note: 0x8 and 0x9 are very easy to mix up. The correct order is 8 = overvoltage, 9 = undervoltage.

Note: These are the common codes the driver can report. The shipped SDK provides text for only some of them and displays the rest as “unknown error”; interpret them from this table.

If you cannot resolve the problem yourself, record the product model and serial number, a description of the symptom and how often it occurs, and the host operating system version, then contact us.

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4 FAQ

4.1 Q&A

Q: How wide is the gripper aperture? A: The effective stroke is 85.000 mm. The SDK position p runs from 0 at the fully closed end to 85.000 at the fully open end, the whole stroke is usable, and p is the aperture a caliper

would measure. Per finger that is 42.500 mm.

Q: How heavy an object can it grip? A: It depends on your workpiece, not on the gripper alone. Grip is friction holding, so the figure depends on the gripping force, the friction coefficient of the contact surface, and the pose. This manual publishes no load figure; contact us for the data for your application.

Q: How large is the gripping force? A: It is not published. `grasp(force_n=...)` accepts a value, but the SDK converts it through an uncalibrated nominal coefficient, so it is a qualitative reference only and must not be used as a calibrated gripping force.

Q: Can I set the gripping force? A: `grasp(force_n=...)` takes a force, but entering one does not make the force accurate. For a position grasp that stops on contact, use `close()`, which takes no force argument at all.

Q: Is Windows supported? A: No. The development kit currently supports Linux only.

Q: Is it water- and dust-proof? A: The IP rating is not yet certified. Contact us before using it in an environment with dust, moisture, or cutting fluid.

Q: Does the gripper release when power is cut? A: Yes. This product has no power-off self-locking. If your application cannot accept that, you must add a separate mechanical holding device.

Q: Must the 24 V supply be connected? A: Yes. With only USB-CAN connected, it can communicate and report status, but it will not move, and it reports undervoltage. That is normal, not a fault.

Q: How do I stop the gripper? A: The emergency stop must be implemented in hardware: cut the 24 V drive power (note that this product has no power-off self-locking, so the fingers will release when you cut it), or wire an independent emergency-stop circuit outside the host computer.

Q: What is the technical support channel? A: The NEXFORM ROBOTICS technical team. Contact details are in the documents shipped with the product and through the sales channel, or the contact person from your purchase can forward you to technical support directly.

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5 Related documents and support

This manual covers the product's appearance, technical specifications, safety notices, selection guidance, and daily maintenance. Installation, wiring, and the first run are in the Usage Tutorial; software interfaces and protocol details are in the Software Development Manual. The repositories are listed below:

Document / repository	Contents
the Usage Tutorial	Unpacking, mounting, wiring, and the first run
the Software Development Manual	SDK usage, ecosystem integration, and protocol appendix
litegrip-python	Python SDK
litegrip-cpp	C++ SDK
litegrip-ros2	ROS 2 driver

Document / repository	Contents
litegrip-studio	Host application
github.com/nexform-tech	All repositories and document downloads

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End of document If you have questions or need more information, contact the NEXFORM ROBOTICS technical team.